Data-Driven Model Structure Notes

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1 Question 1.

Weight only needs the single subscript i because that is going to remain constant through each sweat measurement, so they only need to account for it on the subject level. The no-tattoo indicator, on the other hand, will change despite the subject being the same, so this is on the observation level.

2 Question 2.

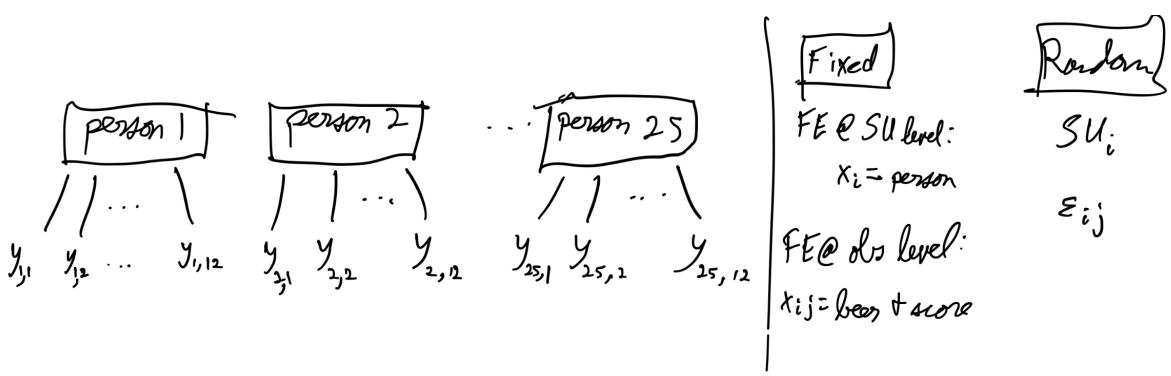
If the classmate used age_{ij} despite it being measured one per subject, at the observation level, *modeldiagramR* uses data to generate a corrected model by creating a model diagram based on the data, which would make this student's mistake clearly visible.

3 Question 3.

I'd use the LearnVizLMM package to check the model diagram fro lab 7 before cleaning and wrangling the data because it doesn't require the data or a model object to draw the diagram.

4 Question 4.1

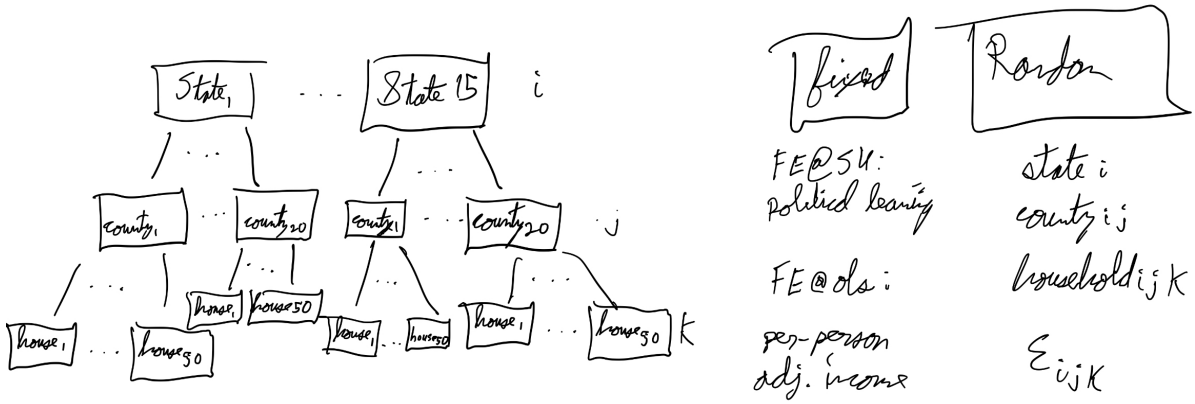
Measurement structure:



- fixed effects column:
  - people @SU level (x_i)
  - Time they took quiz, liters of beer consumed @obs level (x_{ij})

- random effects column: $\mathrm{SU_i}$, ϵ_{ij}
- i = 1-25 (index people)
- j = 1-12, (index observation within people (number of tests taken))

Theoretical model: $\mathrm{IQScore}_{ij} = \mu_{ij} + Person_i + \epsilon_{ij}$
$\mu_{ij} = \beta_0 + \beta_1 I_{test2} + \ldots + \beta_1 1 I_{test12} + \beta_{13}$



- i = 1-15 (index state)
- j = 1-20 (index county)
- k = 1-50 (index households)

Theoretical model: $diff_{ijk} = \mu_{ijk} + state_i + county_{ij} + household_{ijk} + \epsilon_{ijk}$